MD MUJAHIDUL ISLAM MRIDHA

Software Engineer | Full-Stack Development | AI/ML

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SUMMARY

Software Engineer and AI/ML Researcher experienced in production full-stack systems, distributed backends, mobile applications, cloud infrastructure, and applied ML. Strong across TypeScript/JavaScript, Python, React/Next.js, NestJS/FastAPI/Laravel, databases, DevOps, RAG, and computer vision.

TECHNICAL SKILLS

- **Languages:** TypeScript, JavaScript, Python, PHP, C, SQL, MQL, RISC-V Assembly
- **Frontend & Mobile:** Next.js, React, Astro, React Native, Expo, Tailwind CSS, Zustand, TanStack Query
- **Backend & APIs:** NestJS, Node.js, FastAPI, Laravel, REST APIs, JWT, RBAC, HMAC, BullMQ
- **AI/ML & Data:** PyTorch, TensorFlow, LangChain, Ollama, Hugging Face Transformers, OCR, RAG, ChromaDB, MongoDB, PostgreSQL, Redis, DragonflyDB
- **Cloud & DevOps:** Docker, Kubernetes, GitOps, FluxCD, Cloudflare Workers, S3/MinIO, GitHub Actions, Railway
- **Integrations:** SSLCommerz, EPS, BDApps, Appwrite, OneSignal, SMS gateways, payment gateways, Remotion

WORK EXPERIENCE

Software Engineer — *Antimattr Labs* July 2025 – September 2026

- On **QuizBoxBD,** worked across the full stack on backend business logic, UI implementation, and data analysis tools — leading UI and backend architecture reworks, bKash/BDApps/EPS integrations, bug fixing, and end-to-end delivery; built Appsmith back-office workflows, caching for live-leaderboard datasets, and atomic writes for quiz and payment flows to preserve consistency under concurrent load; managed deployments with Docker, Kubernetes, and GitOps (FluxCD).
- On **AmarDeen,** used TanStack Query for data fetching and caching and Zustand for global state management; fixed bugs in the habit tracker, developed the nearest-masjid feature, and implemented client-side data caching, OneSignal notifications, and an audiobook pipeline using multiple TTS technologies.
- On **Let's Math,** built a domain-grounded RAG system for LLM responses, developed an AI agent and tool chain for structured tool calls to generate math-explanatory videos, and synchronized generated narration with video output using Remotion.

Software Engineer — *Fleeting Trails* Jun 2024 – Jun 2025

- Built the **Gulshan Club Hall Booking platform** solo across frontend, backend, database, and admin/back-office, including SSLCommerz payments, booking/availability, reporting, analytics, accounting, and business operations.
- Worked on **Symamlaw**, a real-time law education platform, including frontend, backend development and load-balancing for concurrent quiz traffic and live sessions.

SELECTED PROJECTS

- **QuizBoxBD**: Multi-tenant trivia and gamification platform with tournaments, live leaderboards, and multilingual support.
- **AmarDeen**: Islamic lifestyle app with Quran reader, prayer tools, habit tracking, tasbeeh, and Qibla compass.
- **Let's Math**: AI math-learning platform that generates narrated, animated video explanations for curriculum topics.
- **Gulshan Club Hall Booking**: Banquet management system handling hall booking, availability, reporting, and accounting.
- **Symamlaw**: Law learning platform offering live/recorded classes, courses, exams, and digital resources.

EDUCATION

BRAC University, B.Sc. in Computer Science 2021 – 2026

Dr. Mahbubur Rahman Mollah College, HSC, 2020 | Narayanganj Ideal School, SSC, 2018

REFERENCES

Ariyan Jahangir — CEO, Fleeting Trails | ariyanjahangireng@gmail.com

Navid Anindya Neel — Technical Lead, Antimattr Labs | navidanindya@gmail.com